

11.4. All Templates Used for Indirect Scalar Adjective Intensity Ranking Probing

Index	weak-strong	Index	strong-weak
0	ADJ _{weak} but not ADJ _{strong}	22	not ADJ _{strong} just ADJ _{weak}
1	ADJ _{weak} and almost ADJ _{strong}	23	not ADJ _{strong} , just ADJ _{weak}
2	ADJ _{weak} and even ADJ _{strong}	24	not ADJ _{strong} but just ADJ _{weak}
3	ADJ _{weak} or even ADJ _{strong}	25	not ADJ _{strong} , but just ADJ _{weak}
4	ADJ _{weak} although not ADJ _{strong}	26	not ADJ _{strong} but still ADJ _{weak}
5	ADJ _{weak} if not ADJ _{strong}	27	not ADJ _{strong} , but just ADJ _{weak}
6	ADJ _{weak} though not ADJ _{strong}	28	not ADJ _{strong} still ADJ _{weak}
7	ADJ _{weak} or almost ADJ _{strong}	29	not ADJ _{strong} , still ADJ _{weak}
8	ADJ _{weak} , and even ADJ _{strong}	30	not ADJ _{strong} although still ADJ _{weak}
9	ADJ _{weak} , or even ADJ _{strong}	31	not ADJ _{strong} , although still ADJ _{weak}
10	ADJ _{weak} , or almost ADJ _{strong}	32	not ADJ _{strong} , though still ADJ _{weak}
11	ADJ _{weak} , and almost ADJ _{strong}	33	not ADJ _{strong} though still ADJ _{weak}
12	ADJ _{weak} though not ADJ _{strong}		
13	ADJ _{weak} , although not ADJ _{strong}		
14	ADJ _{weak} , but not ADJ _{strong}		
15	ADJ _{weak} , if not ADJ _{strong}		
16	not only ADJ _{weak} but ADJ _{strong}		
17	not just ADJ _{weak} but ADJ _{strong}		
18	ADJ _{weak} even ADJ _{strong}		
19	ADJ _{weak} almost ADJ _{strong}		
20	ADJ _{weak} , even ADJ _{strong}		
21	ADJ _{weak} , almost ADJ _{strong}		

Table 12: All templates used in computing scalar adjective intensity ranking.

11.5. Held-out Prompt Engineering Results

In the main content of the paper, we report indirect probing results using the best-performing template for each model. This may incur some concerns about generalizability. Here, we choose the best-performing prompt in the other two datasets when evaluating one dataset. For GPT-4, we simply use the best template found in other models. Our general conclusion remains unchanged.

Models	DM	CD	WK
Google Ngram	0.287 ₁	0.075 ₁	0.368 ₁
BERT-b	0.425 ₁	0.066 ₁	0.167 ₁
BERT-I	0.482 ₃	0.102 ₁	0.250 ₁
RoBERTa-b	0.266 ₁	0.050 ₁	0.114 ₁
RoBERTa-I	0.463 ₁	0.100 ₁	0.286 ₁
Falcon	0.265 ₁	0.045 ₁	0.167 ₁
GPT-4	0.540	0.273	0.500
Flan-T5-xl	0.544 ₃	0.177 ₃	0.500 ₃
Flan-T5-xxl	0.515 ₃	0.156 ₃	0.364 ₃

Table 13: Indirect probing results for scale membership with the best prompt found in held-out datasets. The best results per dataset are in bold. Subscripted digits are the best template indexes.

Models	DM	CD	WK
Google ngram	0.312 ₀	0.222 ₀	0.442 ₀
BERT-b	0.546 ₁	0.509 ₁	0.623 ₂₁
BERT-I	0.529 ₂₁	0.518 ₃₂	0.738 ₃₂
RoBERTa-b	0.544 ₁₄	0.527 ₁₉	0.689 ₁₄
RoBERTa-I	0.513 ₄	0.555 ₄	0.721 ₆
Falcon	0.451 ₁₈	0.567 ₁₈	0.607 ₁₈
GPT-4	0.666	0.739	0.852
Flan-T5-xl	0.648 ₁₆	0.612 ₁₆	0.754 ₁₇
Flan-T5-xxl	0.564 ₁₇	0.582 ₁₆	0.770 ₁₇

Table 14: Indirect scalar intensity ranking table. Subscripted numbers are the best-performing template number. The best results across models are in bold.

11.6. Model Implementation Details

Flan-T5 and Falcon models are run with float16 precision. All open-source models are run on V100, A100, and M1 chip. GPT-4 is queried via OpenAI API in 2023. Temperature is set to 0 and top_p is 1 where applicable.

11.7. Prompts Used for GPT-4

In this section, we provide two prompt examples for GPT-4.

Adjective scale alignment *Do not provide explanations. Give five most likely words following the phrase: ADJ_{weak} or even*

Adjective intensity ranking *Prompt: Do not provide explanations. Which of the following phrases is more natural? Answer none if they are equally unnatural. A. not just ADJ_{weak} but ADJ_{strong} B. not just ADJ_{strong} but ADJ_{weak} .*

In adjective intensity ranking, we randomly shuffle the correct answer index to avoid heuristics. The model is expected to return the first phrase as the answer for adjective pairs with unequal intensities, and none for those with equal intensities.

11.8. Scalar Diversity Results for All Non-baseline Models (except GPT-4) and Strategies

Strategy	Datasets	Falcon	Flan-T5-xl	Flan-T5-xxl
sy	RX	0.458	0.714	0.897
	GZ	0.421	0.683	0.850
	PVT	0.425	0.688	0.689
	Average	0.435	0.695	0.812
wy	RX	0.135	0.714	0.855
	GZ	0.323	0.683	0.850
	PVT	0.206	0.688	0.726
	Average	0.221	0.695	0.811
cy	RX	0.281	0.835	0.855
	GZ	0.534	0.757	0.867
	PVT	0.578	0.746	0.726
	Average	0.464	0.779	0.816

Table 15: Full table of scalar diversity results. The best results per model per dataset are marked in bold.